Assignment No. 2

BT612T – Fermentation Technology (Theory)

By M.Kashif(03064963933)

Spring 2024

Total Marks: 10
Due date: 4-07-2024

- To solve this assignment, you should have good command over lectures 38-47.

Instructions:

1. This is an in-line submission format.
2. You have to type your answer in the LMS.
3. Copy and paste option is disabled.
4. Once submitted you would not be able to edit it.
5. Give the answer according to question only and avoid irrelevant details.
6. Email submission is not allowed in any case.

Please note that your assignment will not be graded if:

- It is submitted after due date
- Cheating or copying of assignment is strictly prohibited. The cheated or copied assignment will be marked '**Zero**'.

Question No. 1

How does pH affect the microbial growth and product formation in fermentation?

pH is a critical factor that affects microbial growth and product formation in fermentation processes. Here's a concise overview of its key impacts:

1. **Enzyme activity**: pH influences the activity and stability of enzymes involved in microbial metabolism and product formation.
2. **Nutrient uptake**: The pH affects the ionization of nutrients, impacting their availability and uptake by microorganisms.
3. **Cell membrane function**: Extreme pH levels can disrupt cell membrane integrity, affecting nutrient transport and cellular functions.
4. **Growth rate**: Most microorganisms have an optimal pH range for growth. Deviations from this range can slow growth or inhibit it entirely.
5. **Product formation**: pH can influence metabolic pathways, potentially shifting the balance between desired and undesired products.

by M.Kashif(03064963933)

6. **Contamination risk**: Maintaining a specific pH can help control contamination by creating an unfavorable environment for unwanted microorganisms.

7. **Stress responses**: pH fluctuations can trigger stress responses in microorganisms, potentially affecting their metabolism and product formation.

Question No. 2

Design a hypothetical fermentation media recipe for the production of alpha amylase including an inducer. Justify your choice of inducer.

Ans: (3)

Fermentation Media Recipe for Alpha-Amylase Production

Base Media Components (per liter):

- Glucose: 10 g
- Yeast extract: 5 g
- Peptone: 5 g
- KH_2PO_4: 2 g
- $MgSO_4 \cdot 7H_2O$: 0.5 g
- $CaCl_2 \cdot 2H_2O$: 0.1 g

Trace Elements Solution (add 10 mL/L):

- $FeSO_4 \cdot 7H_2O$: 100 mg
- $MnSO_4 \cdot H_2O$: 100 mg
- $ZnSO_4 \cdot 7H_2O$: 100 mg
- $CuSO_4 \cdot 5H_2O$: 10 mg

Inducer:

- Maltose: 5 g/L

pH:

Adjust to 6.5 using NaOH or HCl

Sterilization:

Autoclave at 121°C for 15 minutes

Justification for Inducer Choice:

Maltose is selected as the inducer for alpha-amylase production for the following reasons:

1. **Substrate mimicry**: Maltose is a disaccharide composed of two glucose units linked by an α(1→4) bond. This structure mimics the bonds that alpha-amylase cleaves in starch molecules, making maltose an effective inducer.

2. **Catabolite repression avoidance**: Unlike glucose, maltose doesn't cause strong catabolite repression. This allows for better induction of alpha-amylase genes even in the presence of other carbon sources.

by M.Kashif(03064963933)

3. **Gradual induction**: Maltose provides a more gradual and sustained induction compared to larger starch molecules, which can be advantageous for consistent enzyme production throughout the fermentation process.
4. **Ease of use**: Maltose is readily soluble and easy to incorporate into the media, making it practical for large-scale fermentations.

5. **Dual functionality**: Maltose can serve both as an inducer and an additional carbon source, potentially enhancing overall biomass and enzyme yield.

Question No. 3

A fermentation process for producing protease using *Bacillus subtilis* is experiencing poor yields due to high level of foam formation. Explain how this issue can be resolved? (3)

Ans

Foam formation is a common problem in fermentation processes, especially when producing proteins like proteases. This issue can significantly reduce yields and process efficiency. Here's how we can address the foam formation problem in your Bacillus subtilis protease fermentation:

1. Antifoam agents:

The most direct approach is to use antifoam agents. These are substances that reduce surface tension and destabilize foam bubbles.

- Silicone-based antifoams: Very effective and commonly used.
- Vegetable oil-based antifoams: Natural alternatives, useful if the product needs to be organic.
- Fatty acid esters: Another effective option.

Add small amounts incrementally to avoid over-suppressing foam, which could reduce oxygen transfer.

2. Mechanical solutions:

- Modify the impeller design or agitation speed to reduce foam formation.
- Install foam breakers or mechanical foam destroyers in the fermenter.

3. Optimize aeration:

- Adjust the aeration rate. Sometimes, reducing air flow can help mitigate foam without significantly impacting oxygen transfer.
- Use microspargers for finer bubbles, which can reduce foaming while maintaining good oxygen transfer.

4. Media optimization:

- Reduce protein-rich components in the media, as these often contribute to foaming.
- Adjust the concentration of minerals, especially divalent cations like Ca^{2+} and Mg^{2+}, which can influence foam stability.

5. Genetic engineering:

- If possible, engineer the B. subtilis strain to reduce production of foam-causing compounds.

6. Process control:

- Implement foam sensors and automated antifoam addition systems.

by M.Kashif(03064963933)

- Adjust pH, as foam stability can be pH-dependent.

7. Fermenter design:
- Ensure adequate headspace in the fermenter.
- Use specially designed foam-control baffles.

Would you like me to elaborate on any of these solutions or discuss how to implement them in your specific process?

----------**GOOD LUCK**----------

by M.Kashif(03064963933